Iftekhirul Islam Muhid

Electrical and Electronic Engineer | Robotics & Autonomous Systems | Machine Learning & Deep Learning
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Education

Gopalganj Science and Technology University March 2022 – October 2026
Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering *CGPA: 3.25 out of 4.0*

Undergraduate Thesis

Computational Investigation of $MgSiP_2$-Based Solar Cells Using DFT, Device Simulation, and Machine Learning

- Performed DFT calculations (Quantum ESPRESSO): band gap of **1.37 eV**, absorption coefficient of $\sim$**10^6** cm^{-1}.
- Optimized the device in SCAPS-1D via OFAT sweeps, reaching a simulated efficiency of **27.48%**.
- Built a two-stage, physics-decomposed ML model on 4,790 samples ($\mathbf{R^2 = 0.9908}$, **RMSE = 0.522**), with SHAP analysis of key parameters.

Publications

- M. Nayem, N. H. Ratul, **I. I. Muhid**, M. Mohiuzzaman, M. S. Farid, S. A. Dipa, "A Reliability Oriented Ensemble Learning Framework for Cardiovascular Disease Prediction with SHAP Based Explainability," IEEE SPICSCON, 2026. **(Accepted)**

Research Interests

Robotics & Autonomous Systems · UAVs & Aerial Robotics · Machine Learning & Deep Learning for Autonomous Systems

Projects

Autonomous Line Following Robot | *Arduino Programming*

- Detects the line in real time with an IR (QTR) sensor array and chooses the correct track dynamically.
- Uses PID control for smooth sharp turns and minimal deviation.
- Built with Arduino Nano, QTR sensor array, and motor driver.

The Health Nourish App | *Python, Flutter, Dart*

- Co-developed a mobile app with a LLM-powered chatbot for personalized diet routines and health tips.

Skills

Programming: Python, C, C++, MATLAB, Dart, Arduino, 8086 Assembly
Machine Learning & Data Science: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly
Design & Automation Tools: MATLAB Simulink, TIA Portal, Proteus, AutoCAD, Logisim
Scientific Computing & Simulation: Quantum ESPRESSO (DFT), SCAPS-1D
Developer Tools: Git, GitHub, Linux, Shell Scripting, LaTeX

Training

Training Institute for Chemical Industries (TICI), BCIC Aug. 2025
Industrial Technology on Electrical Engineering & Instrumentation (2 weeks) *Grade: A+*

- Covered power systems (transformers, generators, substations, earthing, synchronization) and process instrumentation & automation (control loops, PLC, DCS, SCADA).

Awards & Achievements

Pre-Seed Funding Award, BDT 25,000 | *Team EduVeno* 2025

- Led the EduVeno team in securing BDT 25,000 in pre-seed funding through the University Innovation Hub Programme (UIHP), Cohort 5, GSTU.

IEEEXtreme 17.0 2023

- Ranked 17th position in the Bangladesh region.

Leadership & Teaching

- President, Programming and Soft Skills Club (2025 – 2026)
- Webmaster, IEEE GSTU Student Branch (2024 – 2025)
- Founder & Director, The Wizzy Corp. (A start-up company, founded in 2024)
- Instructor of C and Python Programming at the University Programming Club
- Organized and mentored participants of RoboFest'24 (Intra-University Robotics Competition)
- Participated in national robotics competitions such as Ignition and BITROBOT

References

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